
Redefining Academic Integrity in the Age of Generative Artificial Intelligence: The Essential Contribution of Artificial Intelligence Ethics

ANDRÉANE SABOURIN LAFLAMME and FRÉDÉRIC BRUNEALT

The widespread adoption of generative artificial intelligence (AI) tools has profoundly impacted higher education, reshaping how knowledge is created, shared, and assessed. While these technologies offer new opportunities for teaching and learning, they also introduce significant challenges, particularly in maintaining academic integrity. This article contends that the traditional focus on preventing plagiarism and misconduct is insufficient in the context of generative AI. Instead, it advocates for a renewed culture of academic integrity that incorporates comprehensive AI ethics training across the academic community. This approach goes beyond simplistic prohibitions to address the broader ethical, social, technical, and normative issues that underpin academic integrity in the use of AI tools. By fostering awareness and competence in AI ethics, it seeks to equip students, educators, and administrators with the skills to critically engage with AI technologies, promote fairness, transparency, and accountability, and ensure responsible and ethical use. The article ultimately supports a pragmatist approach to AI ethics, emphasising reflective, contextual, and autonomous decision-making as essential to navigating the complexities of AI in higher education.

Keywords: academic integrity, AI ethics, ethical competence, generative AI, higher education

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ACADEMIC INTEGRITY AND ARTIFICIAL INTELLIGENCE:
BEYOND AIGIARISM

Because ChatGPT was launched by OpenAI in the fall of 2022, followed by the proliferation of competing language models—such as Anthropic’s Claude, Google’s Bard, and, more recently, Gemini—we have seen profound and varied effects in our various spheres of activity. While the use of these new artificial intelligence (AI) tools opens unprecedented possibilities and is already significantly transforming certain practices, we are only just beginning to appreciate the scale of the social repercussions of the now widespread use of these powerful technological tools (Marchildon et al., 2023). Higher education is one of the sectors in which the impacts of recent advances in generative AI are most strongly felt (Gagnon, 2023; Provost, 2022). AI has become increasingly prevalent in the education sector over the past decade (Gaudreau & Lemieux, 2020) and is already being used, for example, as part of admissions processes, to generate predictions about students’ chances of success, for tutoring, to automate assessment and feedback, or to personalise or adapt teaching (Zawaki-Richter et al., 2019).

Now, with recent advances in generative AI, the field of possibilities is widening all the more, offering new opportunities. The spectacular improvement in generative AI models in recent years means that it is now possible to mobilise them in the performance of many cognitive tasks taught and assessed in higher education (Anctil, 2023; Dehouche, 2021). The increased accessibility of generative AI tools such as ChatGPT and their use by students as part of their academic work (KPMG Canada, 2023a) enables complex tasks to be carried out in educational contexts, ranging from synthesis, summarisation, textual, visual, audio, and video content generation to programming (Bala & Colvin, 2023). For example, professors can now use language models such as ChatGPT to support them in developing lesson plans, help them design assessments and exercises, generate examples or analogies to illustrate complex ideas, design interactive exercises, or facilitate translation or conversation for students whose mother tongue is not the language of instruction (Baidoo-Anu & Owusu Ansah, 2023; OpenAI, 2023b; Sabzalieva & Valentini, 2023).

These examples, which illustrate the versatility of large language models in pedagogy, represent only a fraction of the different uses that generative AI can be put to in teaching (Gimpel et al., 2023). It should also be

remembered that models for generating visual, audio, and video content offer countless possibilities in education, not to mention the fact that these tools are also used to develop applications for specific educational uses, with more and more new generative AI products created specifically for the educational sector appearing on the market (Holmes, 2023). In fact, professors' and students' access to generative AI tools, with the new possibilities they offer, seems likely to significantly influence the traditional dynamics of teaching and learning (Cardona et al., 2023).

ChatGPT is undoubtedly the generative AI model that has attracted the most interest since its launch, with the application reaching one million subscribers just five days after its launch, and 10 million monthly subscribers after six months (Porter, 2023). According to a survey of 5140 Canadians conducted in the spring of 2023 (KPMG Canada, 2023b), including 222 students (4.3 per cent) attending university, college, vocational or high school, one in five Canadians uses generative AI on a daily basis as part of their work or studies. According to the survey, 52 per cent of Canadians aged eighteen and over have used generative AI models in their schoolwork, to generate ideas (63 per cent), conduct research (53 per cent), write essays and reports, or summarise or synthesise (36 per cent). The survey results also show that there is currently a great deal of confusion surrounding the perception and use of generative AI tools by students as part of their coursework: barely a third (36 per cent) of students disclose to their professors that they have used generative AI tools to do their coursework, while 60 per cent consider its use to be a form of cheating. Furthermore, 60 per cent of students surveyed claim to be unaware of institutional controls on the use of AI at their educational institution. In the context of preparations for a round table bringing together over forty ministers of education from around the world, the United Nations Educational, Scientific and Cultural Organization (UNESCO, 2023) published a study showing that of the 450 schools and universities surveyed, less than 10 per cent had adopted institutional policies or formal guidelines concerning the use of generative AI applications.

It should probably be recalled that long before the commercial launch of the OpenAI's model in 2022, many students were already using AI as part of their academic work, for example, to edit and correct their texts, but the democratisation of generative AI models such as ChatGPT and their new impressive performance means that ignoring the many issues

raised by the massive use of generative AI is now impossible (Anctil, 2023). Indeed, generative AI models can complete entire assignments in a matter of seconds (Ibrahim et al., 2023; with the quality of the results varying, of course, depending on the subject, level or type of assessment in question). GPT-4, OpenAI's subscription-only model, is even more powerful. OpenAI's (2023) technical report reveals, for example, that GPT-4 not only excelled on the entrance exams to the major American law schools but also passed the bar exams of various states with flying colours, not to mention standardised exams in various university disciplines, ranging from medicine to psychology.

If not properly supervised, the use of generative AI by students in their academic work risks compromising the ability to adequately assess the development of skills and the acquisition of knowledge and could even have a harmful effect on the value of the diplomas awarded by higher education institutions (Conseil supérieur de l'éducation [CSE] & Commission de l'éthique en science et en technologie [CEST], 2024). Despite the many possibilities offered by generative AI tools in higher education, the risk that AI may facilitate various forms of academic misconduct, including plagiarism, is the issue that occupies a prominent place in the debate on the use of generative AI in education (Crawford et al., 2023).

Plagiarism: From Repression to Prevention

The question of the intellectual property of content produced by generative AI models is an issue that affects different spheres of society (and this issue is also a legal matter), but it takes on a particular dimension and raises specific issues in the context of higher education and research (Crompton & Burke, 2023). The ability of generative AI models to produce essays or answer university-level exam questions has indeed given rise to lively debate as to whether this use is morally acceptable in the context of academic work or whether it should on the contrary be prohibited (Kurtz et al., 2024; Malik et al., 2023). Faced with the risk of students outsourcing their work to generative AI on a massive scale, some academic institutions and professors, in certain contexts, have decided to prohibit the use of generative AI by students for academic work, associating its use with plagiarism, punishable by heavy penalties (Schwartz, 2023). For instance, some Quebec higher education establishments have updated their institutional policies, associating the use of AI, unless

explicitly authorised as part of an assessment, with fraud or plagiarism, and providing for sanctions accordingly (Barbaro, 2023).

Although at first glance, banning the use of generative AI in certain evaluation contexts may seem a simple and effective solution, its application proves highly complex, if not unenforceable. Indeed, given that language models such as ChatGPT generate new content with every query, establishing beyond doubt that a text has been generated by AI (unless the student admits to it) is virtually impossible, unlike other, more traditional forms of plagiarism (Bussière McNicoll, 2023). Even an educator who knows how to recognise the succinct and regular style of large language models could easily be fooled, as deft rewording can easily mask the distinctive features of an AI-generated text. Recently, various programs designed to identify AI-generated texts, such as Turnitin, GPT Zero or Compilatio, have been launched on the market. Although they are somehow accurate enough to recognise human writing, their ability to identify AI productions is far less effective (Sadasivan et al., 2023), and the problem is that they also generate a large number of false positives (Elkhatat et al., 2023).

Given their limited effectiveness, the use of such tools can lead to unfounded accusations of plagiarism (Edwards, 2023). This can, of course, have serious prejudicial consequences for students. Even if the unfounded accusation is eventually overturned by an appeal committee for lack of evidence, the appeal process is likely to generate significant stress for the student who ends up with the burden of proof. Thus, the impossibility of effectively and accurately detecting AI-generated content seems to compromise the application of the unilateral prohibition on AI use. Unilaterally associating the use of generative AI tools with plagiarism and opting for a total ban on its use may not be the most promising way of ensuring academic integrity in academia (Peters, 2023). Indeed, research into academic integrity concluded as long ago as 2009 (Bretag & Mahmud, 2009) that automated systems for detecting academic fraud would always lag behind technological innovations, a prediction that seems to be borne out in the era of generative AI.

Faced with the difficulty, if not the impossibility, of effectively and reliably detecting AI-generated content, it seems that Quebec's higher education establishments are tending to abandon the repressive approach (Bussi res-McNicoll, 2023). Contrary to what might have been predicted

by the prohibitive wave that swept in the wake of the launch of the first generative AI models (Schwartz, 2023), more than a year later, few Quebec higher education establishments have modified their institutional policies to include a unilateral ban on AIgism (Bussière-McNicoll, 2024). While this appears to be a wise stance, given the difficulties outlined earlier, we maintain that the status quo is nevertheless unsustainable.

Indeed, higher education institutions must work actively to put in place mechanisms, structures, and training to promote a renewed preventive approach to academic integrity in the unprecedented context of the democratisation of generative AI. This posture is in line with a dominant current of thought in academic integrity research that, long before the arrival of generative AI on the market, stipulated that the prohibitive approach to academic integrity was proving unsatisfactory and should give way to a more positive approach based on the establishment of a *culture of academic integrity* (Solmon, 2018). As outlined by Martine Peters (2023), in the context of the democratisation of generative AI, and given the technological limitations of the prohibitive and repressive approach to AI, this perspective on academic integrity is even more relevant. This approach suggests that fostering academic integrity in higher education in this new technological context requires implementing positive and proactive measures to empower all members of the academic community to engage with generative AI tools in an informed, independent, and critical manner, ensuring their responsible and ethical use.

A Renewed Culture of Academic Integrity Based on AI Ethics

We argue that the implementation of such a culture of academic integrity in the era of generative AI must be based on a preventive and broad approach that includes considerations for the wider ethical issues related to the use of AI in academia. More specifically, our thesis is that academic integrity issues cannot be addressed separately from the broader ethical, social, technical, and legal issues in which they are embedded. Therefore, we maintain that renewing the culture of academic integrity in the context of widespread AI use must include adequate training in AI ethics for the entire academic community. As demonstrated in *AI Ethics Training in Higher Education: Competency Framework* (Bruneault et al., 2022), such training in AI ethics should not only provide adequate information on the ethical dilemmas associated with the use of AI in an academic

context but also cover broader topics such as on the risks of erroneous results and misinformation, biases, stereotypes, and prejudices in the results generated by the models (Bommasani, 2021), as well as on the intellectual property and personal data protection issues associated with them (Tsamados et al., 2021).

Promoting academic integrity in this new technological context will also undoubtedly require certain transformations in pedagogical and assessment approaches. Indeed, according to UNESCO (2024), the long-term strategy for institutions and educators will be to rethink the design of written assignments so that tasks required will not be tasks that generative AI tools can perform better than human learners. The democratisation of generative AI tools will therefore also require that we redesign our pedagogical and assessment approaches so that they remain relevant, fair and equitable. Moral philosophy, and, more specifically, AI ethics, can be a powerful ally in the redefinition of our practices, on the appropriate and differentiated frameworks needed for the responsible use of generative AI in higher education, as well as in the exercise of redefining academic integrity demanded by the ongoing technological changes we face.

Academic integrity cannot be reduced to the prohibition of disloyal, dishonest or fraudulent behaviour but should rather be defined as a positive commitment to the following moral values: honesty, trust, fairness, responsibility, respect and courage (International Center for Academic Integrity, 2021). These values are actually moral principles. Because these principles necessarily take on new meanings in the context of the availability of generative AI tools, they therefore need to be redefined. Our claim is that for the members of the academic community to fully embrace these redefined principles, a basic competence in AI ethics is required. However, this assumption of competence cannot be taken for granted, whether among students, teachers or administrators (Bruneault et al., 2022); therefore, adequate training is necessary to ensure a minimal level of competence in AI ethics. To foster a renewed culture of academic integrity in the era of generative AI, this training must go beyond simple rule following. It should empower members of the academic community to critically engage with AI tools and understand their ethical implications, enabling them to make informed decisions that reflect these core values in their academic and professional practices. In the following section, we demonstrate that the pragmatist approach to ethical competence

is particularly relevant in this context and aligns with a more positive approach to academic integrity. We also outline the essential components and practical approaches necessary to develop a comprehensive ethical competence among all members of the academic community.

THE PRAGMATIST APPROACH TO ETHICS

There are several approaches to ethics training and the development of ethical competence, each offering its own perspectives and methods (Bégin, 2014). Regarding ethical competence and AI, first, defining ethical competence in a more general way is necessary. However, defining ethical competence presents difficulties. Several ways of defining ethical competence have been proposed (Bégin, 2003), but, as argued before (Bruneault et al., 2022), the pragmatist approach to ethics seems particularly relevant in this particular context. We will now demonstrate that the pragmatist approach to teaching ethics and developing ethical competence can be particularly useful in the collective redefinition of the moral values that form the pillars of academic integrity. To accomplish this, we must first provide a brief overview of the key characteristics and foundational principles of pragmatist ethics.

The pragmatist definition of training and ethical competence is rooted in the philosophical current of American pragmatism—notably in the seminal work of Charles Sanders Peirce (1877), William James (1907/2003), and, more directly, John Dewey (1920/2004). One of the distinctive features of the pragmatist approach to ethics is that it states that being ethically competent cannot be reduced to knowing and applying ethical theories or behaving in accordance with pre-established rules (Dewey, 1920/2004). In fact, the pragmatist definition of ethics focuses on the necessarily reflexive, autonomous, and practical dimensions of the ethical stance, which requires a critical reflection on the validity of rules or norms, code of ethics, and classical theoretical frameworks in ethics (Boudreau, 2019). Dewey indeed approaches ethics in a spirit of practical experimentation that contrasts sharply with the way moral philosophies usually approach such questions (Cometti, 2010, p. 30).

This definition of competence in AI ethics, which is used in this article, was first outlined in *AI Ethics Training in Higher Education: Competency Framework* (Bruneault et al., 2022). Echoing the work of Lacroix et al. (2017) on these issues, we drew on their pragmatist approach to ethics

training, initially developed in an organisational context, and applied it to AI ethics training in higher education. Our objective will be to demonstrate how the approach to ethical competence developed in this framework is particularly useful for considering the role of AI ethical competence development in promoting a culture of academic integrity within higher education institutions. We will now briefly present the framework proposed by [Lacroix et al. \(2017\)](#) to better understand how it is mobilised in the definition of AI ethical competence we proposed and that can contribute to the renewal of a culture of academic integrity in the era of generative AI.

[Lacroix et al. \(2017\)](#) have defined the pragmatist conception of ethical competence as follows: ‘To be ethically competent is to be able to act in ethical situations autonomously and responsibly through the voluntary mobilization of appropriate internal and external resources’ (p. 112, our translation).

Four key features of this approach to ethics training can be identified:

1. It focuses on ‘being competent’ in ethics, rather than merely ‘having competence,’ which emphasises the active, behavioural aspect of ethical practice.
2. It posits that ethical action requires a constant and continuous interaction between the individual and their environment rather than viewing them as separate entities.
3. It highlights the idea of voluntary action, emphasising the conative dimension of ethical competence, ‘willing’ being a fundamental characteristic of ethical action, reflecting autonomy and freedom.
4. It asserts that an ‘ethical situation’ is always grounded in a specific context, which provides the internal and external resources necessary for voluntary action, recognising that ethical problems cannot be fully understood from a detached or decontextualised perspective.

According to [Lacroix et al. \(2017\)](#), this definition of ethical competence implies the development of three essential skills, which are defined as three of the components of ethical competence: ethical sensitivity, reflexive skills, and dialogical skills ([Lacroix et al., 2017](#)).

Being in an Ethical Situation (Ethical Sensitivity)

Drawing on [Dewey’s \(1920/2004\)](#) notion of moral experience, the first component of ethical competence requires the actor’s ability to recognise

when they are experiencing an ethical situation. This experience involves a form of ‘rupture, imbalance, or disturbance in the course of their actions’ (Lacroix et al., 2017, p. 101, our translation). The ability to experience such destabilisation is an essential component of ethical competence, yet it is not as spontaneous or natural as one might think, and ethics training must necessarily involve developing what we can describe as ethical sensitivity. According to Lacroix et al. (2017), only by developing this ability to *be* in an ethical situation can the actor then actively engage in seeking avenues for resolving these situations.

Knowing how to act in Ethical Situations (Reflexive Skills)

This second component of ethical competence involves the expression of a *practical intelligence* in ethical situations. (Lacroix et al., 2017, p. 106). This practical intelligence proceeds from a mobilisation of resources (both internal and external) to adequately understand the ethical situation that challenges the actor and produce an appropriate response to the environment in which the ethical situation emerges (Lacroix et al., 2017, p. 106). To understand and analyse the elements characterising the problematic situation and consider possible solutions to the destabilisation generated by it, the actor must mobilise internal resources that are both conative and cognitive, as well as external resources that are material and social. This process also requires the actor to consciously engage in a dialogue with the problematic situation, actively seeking to transform it through their actions: ‘It is by becoming aware of his own practices and ways of assuming his roles in response to the expectations and norms addressed to him that the actor gives himself the possibility of opening himself up to thoughtful transformations of his ways of responding to ethical situations’ (Lacroix et al., 2017, p. 108, our translation).

Interacting in Ethical Situations (Dialogical Skills)

This brings us to the third component of ethical competence, namely, interacting in ethical situations. Interacting competently in an ethical situation involves the ability to generate agreements with others regarding the desired outcomes and the actions to be taken, as achieving these goals requires collaboration. These actions are guided by socially shared meanings, conventions, and habits, which influence how the individual uses the resources available to them. These actions are guided by socially

shared meanings, conventions, and habits, which influence how the individual uses the resources available to them (Lacroix et al., 2017, p. 109). This capacity relies on three key elements: attention to others, dialogical skills, and a willingness to be accountable for one's actions.

This is particularly important in the case of ethical competence, which directly concerns the relationship with others. Ethical competence requires 1) attention to the experience of the ethical situation of others; 2) the dialogic capacity to stabilise the social environment from which the situation is apprehended; and 3) the willingness to answer—and therefore to be responsible—for one's own actions in relation to others. It follows that 'ethical situations always involve a register of socially constructed notions and concepts, through which claims to guide people's conduct are expressed. It is on the basis of his understanding and use of these notions and concepts that the actor appreciates and constructs what is problematic in the ethical situation, and it is also through these that she can relate to others and initiate with them a course of action aimed at achieving a desired outcome' (Lacroix et al., 2017, pp. 109–110, our translation).

In line with the dynamic concept, the three components of ethical competence interact with each other synchronously, supporting and reinforcing each other. This is why there are three integral components to one and the same ethical competence:

Interacting with others is something that manifests itself as soon as we enter an ethical situation, through the necessary attention paid to others. To be in an ethical situation is, in effect, to already be involved in interactions with others, which will contribute to the destabilization that calls for a new course of action. Acting in an ethical situation is in turn inseparable from these interactions. Both the competent problematization of the ethical situation and the determination of the solutions and results to be achieved imply, from the outset, a dialogic capacity as well as a willingness to answer for one's choices and actions. (Lacroix et al., 2017, p. 112, our translation)

Ethical competence therefore implies being able to combine these different components.

The 3 components of ethical competence

1. **Being in an ethical situation**
Ethical sensitivity
2. **Knowing how to act in an ethical situation**
Reflective skills
3. **Interacting in an ethical situation**
Dialogical skills

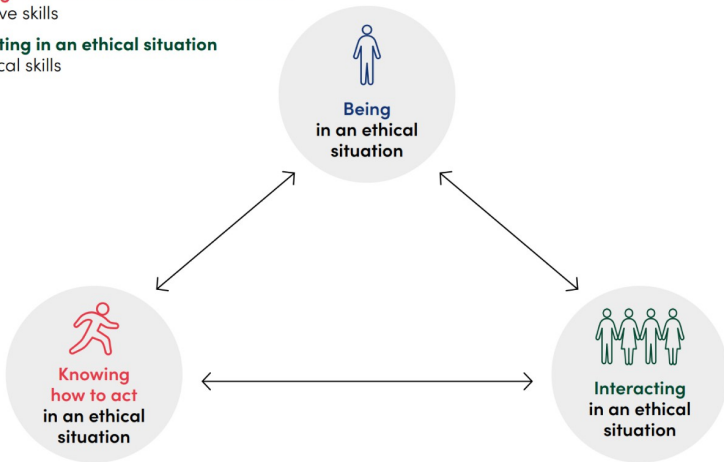


FIGURE 1. The Three Components of Ethical Competence

THE PRAGMATIST APPROACH TO AI ETHICS IN HIGHER EDUCATION

Given the rapid pace of technological development, the emergence of new ethical situations related to the use of AI, and the increasing frequency of our interactions with IA in higher education, the pragmatist approach seems particularly relevant for AI ethics training. This is especially true in the context of the democratisation of generative AI, where these tools are becoming widely accessible and integrated into educational practices.

In this specific context, the three components of ethical competence take on specific meaning. For instance, ethical sensitivity implies that members of the academic community recognise that their own use of generative AI in an academic context may generate situations that raise ethical issues and dilemmas. Reflexive skills require the ability to critically assess and reflect on one's use of generative AI, considering the ethical implications of relying on these tools for learning and teaching. This includes evaluating the appropriateness of using AI for certain tasks, understanding its limitations, and ensuring that its use aligns with broader educational values and objectives. Dialogical skills emphasise the importance of open

and informed dialogue between educators, students, and administrators about the ethical use of generative AI. This dialogue fosters a culture of transparency and shared understanding, enabling all members of the academic community to collaboratively address ethical concerns and develop best practices for AI integration.

This is why, in the Competency Framework for AI Ethics Training in Higher Education (Bruneault et al., 2022, p.12), the definition of ethical competence has been defined as follows:

The competence in AI ethics means being able to act autonomously and responsibly in ethical situations involving AI systems, through the voluntary mobilization of appropriate internal and external resources.

While it is true that these components of ethical competence can be applied in ethics training across various contexts—ranging from bioethics and environmental ethics to business ethics, and, of course, AI ethics—each field of applied ethics involves specific and distinct ethical issues where these three components are deployed. This is why we found it necessary to define four distinct fields of competence in our competency framework (Bruneault et al., 2022). In addition to these three components that we have adapted from Lacroix et al. (2017), we have added four fields of competence in AI ethics. We believe these fields encompass the full range of ethical issues associated with the use of generative AI in higher education and can help establish the foundation for renewing a culture of academic integrity in the era of generative AI in higher education. Starting from the pragmatist definition of ethical competence regarding the use of AI, we now consider how developing a culture of academic integrity in the era of generative AI must be based on the development of these four fields of competence in AI ethics.

The Fields of Competency in AI Ethics

Because the components of ethical competence can only be deployed in the assessment of specific situations, we need to specify the practical field in which these components can take shape. While different fields of application would lead to different fields of competence (e.g., in bioethics, environmental ethics, or business ethics), the competence in AI ethics, now necessary for developing a renewed culture of academic integrity,

requires us to situate the ethical issues linked to this technology first and foremost in the context of the rapid technological development and its impact on individuals and the groups to which they belong.

First, the ethical issues related to AI stem from their specific technical constitution and functioning, which must be considered to fully grasp the extent of the individual and social transformations they bring about. Also, it should be remembered that AI-related ethical issues are part and parcel of moral dilemmas that are not only sometimes new iterations of long-standing moral debates and value conflicts but also sometimes quite new, even unprecedented. It is important to situate the ethical issues of AI in its social context (that of the design, development and deployment of this technology), highlighting the inequalities that characterise this social context and the power relations that underpin it. Finally, we believe that placing the ethical issues of AI within the framework delimited by the norms, specific expectations, and formal obligations that contribute to the normative framing of these technological innovations is essential. This leads us to distinguish four fields of competence in AI ethics linked to the

- a. technical aspects of AI;
- b. moral dilemmas of AI;
- c. socio-technical context of AI; and
- d. normative frameworks for AI.

Technical Aspects of AI

The first field of competence in AI ethics relates to the technical aspects of AI systems (AIS). More specifically, one of the fundamental requirements for developing competence in AI ethics involves having a minimum understanding of the technical processes involved in AIS. The speed of technological development and the equally rapid evolution of the uses of technological innovations require a person competent in AI ethics to constantly update their knowledge of the latest technological devices to anticipate their applications and potential effects (Brey, 2012). At the same time, this competent actor must avoid basing their assessments of these issues on a poor understanding of technical processes or on extrapolations out of step with current technological development, which could lead them to deal primarily with highly hypothetical ethical issues. Such approaches risk being more of a distraction than a

**The 4 fields of competency
in AI ethics**

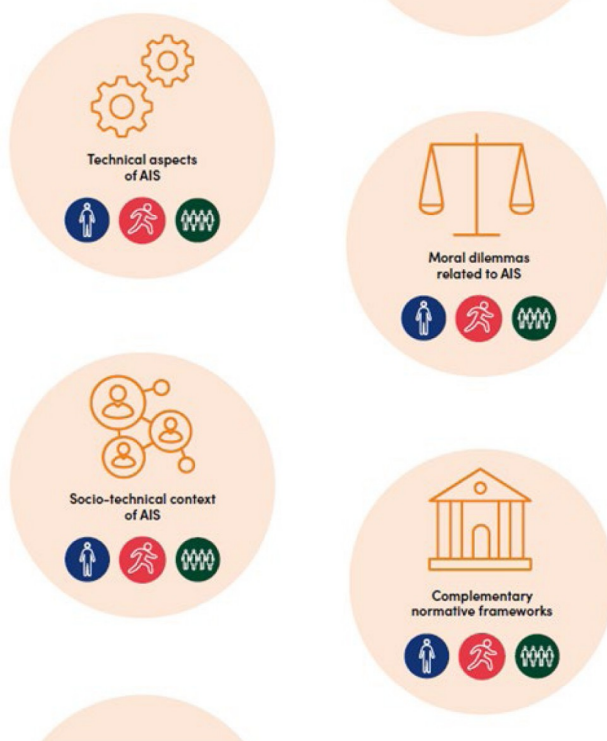


FIGURE 2. The Four Fields of Competency in AI Ethics

consideration of the very real problems raised by the technological devices that currently exist (or can be anticipated in the light of recent developments). While there is undoubtedly a forward-looking element to this assessment of the ethical issues of AI linked to the technical aspects of AIS, it seems that a proper knowledge of these technical aspects helps to avoid directing ethical debates towards purely speculative questions (Richard et al., 2023). This posture is also consistent with the pragmatist approach to ethics, which demands a practical and contextual approach to ethics, thus avoiding the pitfalls of certain overly normative or abstract approaches to ethics (Keulartz, 2002).

However, this first field of ethical competence also implies that the competent actor in AI ethics must be able to appreciate the complexity of

the relationship between human beings and technical objects. Echoing work in the philosophy of technology (Verbeek, 2005; Winner, 1980), approaching technology by questioning the idea that it is neutral is essential, since it is merely a set of tools available to human beings, who can use them as they see fit. The relationship that human beings have with technical objects influences their relationship with the world, and the relationship with technical objects is even a fundamental component of what it is to be human.

To contribute to the development of a culture of academic integrity in the context of the democratisation of tools based on generative AI, members of the academic community need to be trained in the basic workings of these highly complex AI systems.

The first element of adequate training in AI ethics should emphasise the technical peculiarities of generative AI, in particular the scale of the training data used for this type of system and what this implies for their operation and the quality of their outcome. Moreover, understanding how the technical functioning of generative AI models inherently leads to risks of erroneous outputs is important; this is a key issue related to the risk of misinformation they represent (Helmus, 2022).

Despite their impressive capabilities, generative AI models often produce content that is incorrect, misleading, or not directly based on their training data, a phenomenon sometimes referred to by experts as ‘hallucinations’ or fabrications (Maclure, 2023; Bommasani et al., 2023; Spitale et al., 2023; Weise & Catz, 2023). This issue is particularly concerning in an educational context, where the truthfulness and accuracy of content are vital. These errors stem from the inherent limitations of language models, which function as statistical systems designed to predict the relevance of words in response to a query rather than providing factual or referenced information (CSE & CEST, 2024).

The risks of misinformation become even more alarming given the vulnerability of learners and the apparent confidence with which language models can present incorrect or fabricated content, amplifying the well-known Eliza effect (CEST, 2023), where human traits are attributed to AI systems and, in this case, granting them unwarranted authority on specific subjects. In this first field of competence in AI ethics to be developed with a view to promoting a culture of academic integrity, it is thus important to highlight the particularities of a technical device

that is able to generate novel content (often in the form of a “conversation” with a conversational robot) and the implications of this new type of technical object on human relations and the representations we have of ourselves.

Moral Dilemmas of AI

The second field of expertise in AI ethics concerns the moral dilemmas raised by AI systems. These moral dilemmas are part of conflicts of values and norms that go beyond the strict framework of AI development and have their roots in a long history. The main theories in moral philosophy, namely, deontology, virtue ethics, consequentialism and emotivism (Hösle, 2004) and the principles they propose attempt to offer possible solutions to these dilemmas. The tensions between autonomy, well-being (notably in the form of beneficence and non-maleficence), privacy, security, confidentiality, consent, transparency, equity (to name but the most classic examples) are not specific to the development of AI systems, and possible resolutions to these tensions have not waited for the development of AI to be put forward. The competent actor in AI ethics will, however, be able to grasp the way in which AI-related technological innovations represent a new iteration for the expression of these conflicts.

To do this, the competent actor will need to develop a general understanding of the main theoretical approaches in moral philosophy, the tensions between them and their limits, and the particularities of the issues at stake in AI ethics, which may require the reconceptualisation of certain principles derived from classical ethical frameworks (van den Hoven, 2010). Furthermore, the competent actor in AI ethics will also need to be able to grasp the particularities of ethical situations involving AI systems, or even their novelty, by developing their ability to identify the different epistemological, normative and transversal issues at stake in problems related to AI (Mittelstadt et al., 2016). In so doing, they will also be able to foresee the unprecedented conflicts of values and norms implied by using AI generative systems in higher education, particularly regarding the explicability and traceability of these models.

In the academic context, one of the critical challenges posed by using generative AI is commonly referred to as the ‘black box’ problem, which is related to the opaque nature of these systems and their underlying process (Gillani et al., 2023). Generative AI models, such as neural networks,

often operate in ways that are not fully transparent or understandable, even to the developers who design them (Bender, 2021).

The black-box problem underlies in fact a wide range of ethical issues that an adequate education in AI ethics must unravel, such as transparency and accountability and the ability to critically evaluate the outputs generated by these models (Tsamados et al., 2021). Tools based on generative AI are also bound to raise questions about consent, particularly that of the people who produced the content of the training data used to make these systems possible, and about the privacy of the people who use them. To promote the responsible and ethical use of AI in higher education, the institutional culture must promote the free, informed, and ongoing consent of individuals interacting with a generative AI model, including when human-generated content is fed into these models (CSE & CEST, 2024).

In our view, the development of an appropriate culture of academic integrity in the context of the massive use of generative AI therefore requires that all members of the academic community are made aware of these specific ethical issues and are properly trained to address them in their respective practices.

Socio-Technical Context of AI

The third area of competence in AI ethics concerns the socio-technical context of AI systems (Noble, 2018). Properly situating the ethical issues of AI systems within the socio-economic context in which they are embedded, particularly regarding issues related to equity, diversity, and inclusion (EDI), is crucial to grasp the scope of these issues. Because AI systems are likely, as numerous studies have shown (Eubanks, 2019), to reproduce, amplify and automate already existing discrimination, exclusion, and inequalities towards certain groups, the competent actor in AI ethics must be able to mobilise the conceptual and analytical resources needed to integrate this dimension of ethical issues into their reflection and action. In doing so, they must also understand the ethical issues at stake in the power relations that lie at the heart of the design, development, and deployment of AI systems and situate them in relation to the political dynamics in which they are embedded (Floridi, 2023).

Having a comprehensive understanding of the social and political impacts of tools based on generative AI is therefore becoming an inescapable

aspect of developing a culture of academic integrity at a time when generative AI is becoming increasingly available, particularly because these systems produce results tinged by biases that can generate discrimination. In this regard, it is important to state that AI models are not neutral (Marchildon et al., 2023). The production of biased content by generative AI models poses a significant issue for academia, as these systems often reproduce and amplify the prejudices and stereotypes embedded in their training data (Bender et al., 2021; Buolamwini & Gebru, 2018; Nicoletti & Bass, 2023).

One must also acknowledge that different language models may exhibit distinct political stances and therefore are likely to generate politically oriented content (Rozado, 2023). Also, the content generated by the same model can vary depending on factors such as the language used to perform a query (Zhuo et al., 2023). This can result in biased outputs that may contribute to various forms of exclusions, further automating and exacerbating existing socio-historical inequalities (Marchildon et al., 2023). Generative AI systems can produce results that potentially discriminate against certain individuals, groups, or communities, particularly women (Conseil du statut de la femme, 2023).

Despite ongoing efforts to mitigate these issues, such problems—well known in other areas of AI, such as recommendation systems—also emerge in generative AI models (Nicoletti & Bass, 2023). The reproduction of stereotypes is particularly concerning in academic contexts, where the goal is to promote critical thinking, diversity, and equity and actively combat prejudice and exclusion (Marchildon et al., 2023).

In this respect, while some of these tools can contribute to greater inclusion (UNESCO, 2019), particularly of non-native speakers of the language of instruction, they can also undermine the principles of equity, diversity and inclusion (Collin & Marceau, 2023). Given the performance of generative AI tools, those with greater access to and proficiency in these technologies could gain significant advantages, while less privileged students or those with less developed digital literacy may not be able to benefit equally. This could exacerbate the phenomenon commonly known as the ‘digital divide,’ which refers to the gap between individuals who have access to digital technologies and know how to use them effectively, and those who do not, or who lack the skills to use them optimally, raising important issues of equity (CEST, 2023). This disparity may be

further intensified by the fact that the most powerful generative AI models are often available only through paid subscriptions, creating additional barriers for those who cannot afford them.

What is more, these systems are often offered by digital giants (GAFAM), who have their own economic and political interests that need to be taken into account if they are to be used wisely (Durand-Folco & Martineau, 2024). However, the proliferation of tools based on generative AI in higher education can contribute to accentuating the digital divide (Collin & Marceau, 2023). Educating members of the academic community on these issues is essential to encourage responsible and ethical use of tools based on generative AI.

A culture of academic integrity in the era of the democratisation of generative AI cannot overlook these socio-technical issues and must include ethics training that raises awareness among members of the academic community about these concerns. Such training should specifically address the risks of bias, exclusion, and the reproduction of stereotypes associated with AI use, equipping students, educators, and administrators with the tools to critically engage with AI technologies in ways that promote fairness, transparency, and accountability within the academic environment

Normative Frameworks for AI

The fourth field of competence in AI ethics involves the awareness and the critical assessment of the existing regulatory frameworks for AI, whether these frameworks are legal, professional, or ethical.

The competent actor in AI ethics must be familiar with and be able to compare the main normative frameworks derived from charters and declarations in AI ethics that have been developed in academic circles, by international organisations, and by major industry players. It is also important to be able to identify the obligations prescribed by codes of ethics and other relevant professional normative frameworks (Jobin et al., 2019). Furthermore, it is necessary to contrast the requirements of AI ethics charters and declarations by situating them within the framework of state and international law, particularly regarding the protection of privacy and personal data (Castet-Renard & Eynard, 2023).

The use of large language models poses significant privacy risks (Li et al., 2023). These risks arise, in part, because the training data for these

models often includes personal and sensitive information that can inadvertently be reproduced in query responses. Even when generative AI is not directly fed with personal data, there remains the possibility that the model could infer or deduce sensitive information. Additionally, users may inadvertently share personal details, sensitive data, or private conversations in their queries, leading to confidentiality breaches (Kshetri, 2023). The information contained in user queries can also be used to further train generative AI models, raising further ethical concerns. Plagiarism detection models, for example, present a unique challenge, as students' work is often transferred to external AI systems without their consent. Moreover, when students incorporate pedagogical content created by their professors into their queries, it can create complications regarding intellectual property rights over that content.

To ensure academic integrity and foster the development of a culture of academic integrity adapted to the unprecedented context of the availability of generative AI, it seems necessary to train members of the academic community on these issues but even more so to communicate to them through this training clear institutional guidelines that mark out the use of generative AI.

Furthermore, it should be recognised that these guidelines alone, like other regulatory documents aimed at regulating AI, cannot rule on all possible uses and therefore cannot replace the development of a culture of responsible and ethical use of AI. Adequate training of the academic community in this field of competence in AI ethics could be a valuable asset to adapt pedagogical and evaluative approaches in the context of the availability of tools based on generative AI.

Distinguishing these different fields of competence and the different expectations associated with each of these fields is useful, but it must again be emphasised that this does not at all imply that these fields of competence operate (or even can be analysed) independently of one another. On the contrary, a particular ethical situation involving AI systems will therefore require that the person competent in AI ethics be able to mobilise the appropriate resources, particularly in interaction with others, resources possibly linked to the four fields of competence in AI ethics (and this according to the three components of competence described earlier) to hope to see, analyse, understand, and seek adequate solutions to the problems generated by this situation.

To effectively promote a culture of academic integrity in the era of generative AI, it is crucial to recognise that the three components of ethical competence—being in an ethical situation (ethical sensitivity), knowing how to act in ethical situations (reflexive skills), and interacting in ethical situations (dialogical skills)—must be deployed across the four fields of competence in AI ethics (Bruneault et al., 2022). These fields include the technical aspects of AI, the moral dilemmas of AI, the socio-technical context of AI, and the normative frameworks for AI. Integrating these components within each field ensures a comprehensive approach to AI ethics, fostering an informed, responsible, and ethical use of AI in higher education.

CONCLUSION

The democratisation of the use of generative AI models like ChatGPT, Claude, Bard, and Gemini has brought about profound changes in higher education, reshaping how knowledge is created, shared, and evaluated. As highlighted in the introduction, while these tools open up new possibilities and significantly transform educational practices, they also give way to a variety of challenges. With the increasing use of generative AI tools by students and educators alike, the debate often focuses narrowly on issues like plagiarism, overshadowing the broader ethical, social, technical, and normative concerns that are deeply embedded in the use of AI in academia and that underlie questions related to academic integrity.

The profound and numerous effects of generative AI tools are already having a significant impact on higher education. Yet the potential for AI to transform educational practices and the challenges it presents to academic integrity demand a proactive yet flexible and contextual approach. We have demonstrated that the pragmatist approach to ethics aptly meets these requirements, as it goes beyond mere compliance with rules or prescriptions—although clear rules are essential—and demands an autonomous and contextual reflection by the members of the academic community regarding their own use of generative AI.

Our thesis contends that renewing the culture of academic integrity in this new technological context requires moving beyond repressive measures against academic misconduct, such as AI-based plagiarism or AIgiarism. Instead, it demands a comprehensive approach that integrates AI ethics training across the academic community. Such training must

provide an understanding not only of the risks related to AI use, such as misinformation, bias, and the reproduction of stereotypes, but also of the implications for equity, diversity, and inclusion in educational settings.

As we have argued, fostering an academic culture that embraces the responsible and ethical use of AI must address the technical limitations and possibilities of these systems, the specific moral dilemmas they pose, the social and political dynamics they influence, and the relevance of existing regulatory frameworks. This requires a shift towards a more positive, proactive stance on academic integrity—one that empowers all members of the academic community to critically engage with AI tools, understand their ethical implications, and make informed decisions.

The thesis defended in this article is that the necessary renewal of a culture of integrity in the era of generative AI must be based on the development of competence in AI ethics and that the pragmatist approach to ethics makes defining more precisely the components of this competence and the fields in which it must be deployed possible. The promotion of an academic culture that fosters the honest, ethical, and responsible use of AI cannot do without a discussion of the technical possibilities and limits of these systems, the specific moral dilemmas associated with them, the issues of equity and power they imply, and the relevance of the normative frameworks that guide their use.

To promote a culture of academic integrity in the era of generative AI, we therefore believe that training in AI ethics is essential for the whole academic community and the implementation of the educational mission of higher education institutions. By providing all members of the academic community with the knowledge and skills to navigate the ethical landscape of AI, we can move beyond simplistic prohibitions and towards a more informed and autonomous use of these technologies.

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